ZOO302 Current Solved

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Q.1 Explain Stomach in Ruminants? **Long r**

• Ruminant mammals are the animals that “chew their cud,’’ such as cows, sheep, and deer show some of the most unusual modifications of the stomach, but can chew the food at a more comfortable or safer location.

• The ruminant stomach provides an opportunity for large numbers of microorganisms to digest the cellulase walls of grass and other vegetation.

• Cellulose contains a large amount of energy; however, animals generally lack the ability to produce the enzyme cellulose for digesting cellulose and obtaining its energy.

• In ruminants, the upper portion of the stomach expands to form a large pouch, the rumen, and a smaller reticulum.

• The lower portion of the stomach consists of a small antechamber.

Q.2 Digestion in grasshopper? **r**

• The grasshopper is a representative insect with a complete digestive tract.

• In grasshopper (or cockroach) the digestion of food starts in the mouth. The saliva contains some enzymes. Saliva also helps in lubrication and softening of food. This partially digested food passes through the esophagus into crop where it is stored temporarily. Then, the food passes into gizzard. From gizzard, the food goes into stomach. In the stomach, the enzymes secreted by

elongated glands called hepatic caeca get mixed with the food to digest it completely.

Q.3 Osmoregulation defines? r

• **It is defined as** the maintenance of constant osmotic pressure in the fluids of an organism by the control of water and salt concentrations.

• It means osmoregulation governs water and salt balance, and excretion of metabolic wastes.

• In fishes, reptiles, birds, and mammals, the kidneys are the primary osmoregulatory structures.

Q.4 what is digestion?

Digestion is the chemical and/or mechanical breakdown of complex food into diffusible molecules i.e. particles that individual cells of an animal can absorb.

• Starch into glucose.

• Proteins into amino acids.

• Fats into fatty acids & glycerol.

Q.5 what is Predation?

Predation is one of the most sophisticated feeding strategies, since it requires the capture of live prey.

It is an interaction in which one organism kills another for food. The organisms that does the killing for food is the predator and the organisms that is killed for food is the prey.

Examples:

Wolves hunting a deer or moose, an owl hunting the mice

Q.6 Intracellular and Extracellular digestion? Long r

Intracellular digestion	**Extracellular digestion**
In intracellular digestion, the breakdown of food materials into small molecules occurs inside the food vacuoles within the cell.	In extracellular digestion, the breakdown of food materials into small molecules occurs outside the cell.
In intracellular digestion breakdown of food occurs within the cells. First food in taken into the cell vacuole The cytoplasm secretes digestive enzymes into the food vacuole. Enzymes break larger molecules into smaller ones, which enter the cytoplasm.	In extra cellular digestion, the enzymes are secreted outside the cell into the gut cavity or lumen where digestion occurs. Nutrients from the food then pass into body cells for energy metabolism or biosynthesis
In intracellular digestion, only chemical digestion takes place.	In extracellular digestion, both chemical and mechanical digestion occurs.
It is a simple mechanism of digestion that involves vesicles.	It is a complex mechanism of digestion that involves the organs and glands.

It mainly occurs in protozoans.	It follows in bacteria, fungi, and animals with the alimentary canal

Q.7 Gallbladder function?

• The gallbladder is a small bag near the liver.

• The gallbladder stores the greenish fluid.

Its main function is **to store bile**. Bile helps your digestive system break down fats.

Gallbladder is a small hollow organ where bile is stored and concentrated before it is released into the small intestine.

• The structure and position of the gallbladder can vary significantly among animal species. Because of the importance of bile in fat digestion, the gallbladder is relatively large in carnivores and vertebrates.

Q.8 Temperature Regulations in Fishes? **Long r**

The temperature of the surrounding water determines the body temperature of most fishes.

How Fishes live in extremely cold water?

1. They have “antifreeze” materials in their blood in the form of Polyalcohols (e.g., sorbitol, glycerol) or water soluble peptides and glycopeptides.

• Because these substances lower the freezing point of blood plasma and other body fluids.

2. These fishes also have proteins or protein-sugar compounds that inhibit the growth of ice crystals that begin to form.

3. These adaptations enable these fishes to stay flexible and swim freely in a supercooled state (i.e., at a temperature below the normal freezing temperature of a surrounding water).

4. These fishes have major blood vessels just under the skin.

• Branches deliver blood to the deeper, powerful, red swimming muscles, where smaller vessels are arranged in a countercurrent heat exchanger called the **rete mirabile.**

5• The heat generated by red muscles is not lost but it is transferred in the rete mirabile from venous warm blood to cold arterial blood.

• Some active fishes maintain a core temperature significantly above the temperature of the water.

For example: Bluefin tuna and the great white shark.

6 This arrangement of blood vessels enhances vigorous activity by keeping the swimming muscles several degrees warmer than the tissue near the surface of the fish. • This system has been adaptive for these fishes.

Q.9 Functions of Liver in vertebrates?

• Bilirubin (“red bile’’) and biliverdin (“green bile’’).

• Liver detoxifies various metabolites.

• Liver synthesizes proteins and produces bio chemicals necessary for digestion and growth.

• Destruction of RBCs and formation of bile pigments.

• Protein metabolism

• Synthesis of plasma proteins
• Enzyme synthesis.
• Conversion of ammonia to urea.

Q.10 Antiperistalsis?

• The reverse peristalsis is called Antiperistalsis, which occurs occasionally.
• These movements cause the passage of food from the intestine back into the stomach or even into the mouth. This is also called vomiting.
• During peristalsis the food advances through the GI tract when the rings of circular smooth muscle contract behind it and relax in front of it.

Q.11 what is Gastrointestinal Motility? GI long

• As with any organ, the function of the gastro intestinal tract is determined by the type of tissues it contains.
• Most of the mammalian gastro intestinal tract has the same anatomical structure along its entire length.
• From the outside inward is a thin layer of connective tissue called the **serosa.**
• The serosa forms a moist epithelial sheet called the **peritoneum.**
• Next are the longitudinal smooth-muscle layer and circular smooth-muscle layer.
• Underneath this muscle layer is the submucosa.
• The submucosa contains connective tissue, blood, and lymphatic vessels.

• The mucosa faces the central opening, which is called **lumen.**

• The coordinated contractions of the muscle layers of the gastrointestinal tract mix the food material with various secretions and move the food from the oral cavity to the rectum.

Q.12 Temperature Regulation in Mammals? **Long r**

• Mammals like birds are also active and behaviorally complex vertebrates.

• They can live in habitats all over the earth because they are endotherms.

• They can maintain body temperatures between 35 and 42° C with metabolic heat.

• Animals in hot climates, such as jackrabbits, have mechanisms (e.g., large ears) to get rid of excess heat from the body surface.

• Marine animals, such as seals and whales, thick pelts and a thick layer of insulating fat called **blubber**.

• In the tail and flippers, which have no blubber, a countercurrent system of arteries and veins helps minimize heat loss.

• Migration to warm climates and hibernation enable many different birds and mammals to survive the harsh winter months.

Q.13 Write nutrition and its types?

Nutrition includes all of those processes by which an animal takes in, digests, absorbs, stores, and uses food (nutrients) to meet its metabolic needs.

A nutrient is a component in foods that an organism uses to survive and grow.

Nutrients may be organic or Inorganic.

Organic nutrients: consist of carbohydrates, fats, proteins (or their building blocks, amino acids), and vitamins.

Inorganic nutrients: These are dietary minerals, water and oxygen.

Q.14 Small Intestine?

• The human small intestine is about 4 cm in diameter and 7 to 8 m in length. The small intestine (small bowel) is about 20 feet long and about an inch in diameter.

• Length of human small intestine is intermediate between the small intestines of typical carnivores and herbivores of similar size.

• It reflects the human's omnivorous eating habits.

The small intestine has three parts:

1. Duodenum– It is about 6 m in length and functions primarily in digestion.

2. Jejunum and

3. Ileum.

Q.15 Describe primitive strike of birds. 3m

How and when primitive streak develop in the birds?

The major structural characteristic of avian, gastrulation is the primitive streak.

• This streak is first visible as a thickening of the epiblast at the posterior region of the embryo.The streak eventually extends 60 - 75% of the length of the area pellucida. • The primitive streak defines the axes of the embryo. It extends from posterior to anterior end. • As cells converge to form the primitive streak, a

depression forms within the streak. This depression is called the primitive groove.

Q.16 How teleosts fishes osmoregulate in fresh and marine water? **Long r**

There seminal study demonstrated that freshwater teleosts extracts NaCl from the environment but marine water teleosts ingests seawater absorb intestinal water by absorbing NaCl and excrete the salt via gill transport mechanisms.

In marine **teleost** osmoregulation, the osmotic loss of water across the gill epithelium, the fish ingests seawater and absorbs NaCl. Urinary water loss is kept to a minimum by production of a low volume of urine that is isotonic to the plasama.

A marine fish has an internal osmotic concentration lower than that of the surrounding seawater, so it tends to lose water and gain salt.

Freshwater teleosts are hyperosmotic to the surrounding solution, so they face osmotic gain of water and diffusional loss of NaCl. Freshwater fish have very efficient kidneys that excrete water quickly.

The internal environment of marine fish is hypo osmotic, while that of a Freshwater fish is hyperosmotic.

Q.17 Function of small intestine?

The small intestine has three parts: the duodenum, jejunum, and ileum. It **helps to further digest food coming from the**

stomach. It absorbs nutrients (vitamins, minerals, carbohydrates, fats, proteins) and water from food so they can be used by the body.